RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Social Media Analysis: User Engagement and Content Performance Insights Using Power BI

AD23632 – FRAMEWORK FOR DATA AND VISUAL ANALYTICS

Submitted by

Name: SUBITSHA V

Register Number: 241501219

Department: Artificial Intelligence and Machine Learning

Submitted to

Ms. AKSHAYA V

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ABSTRACT

The Social Media Analysis project studies user engagement and content performance records using a structured dataset and Microsoft Power BI. The dataset contains 3,000 post-level records covering 498 distinct user IDs, six platforms, five content types, five post categories, and five posting time slots. Each record includes engagement statistics (likes, comments, shares), impressions, engagement rate, response time, a sentiment score, an engagement score, a sentiment label, and a viral post flag.

The Power BI report contains five pages. The first is an overview page with category-level comparisons, likes versus shares, the comments and shares split, and sentiment-by-category comparisons. The second compares platforms by engagement score. The third examines how impressions are shared between viral and non-viral posts across posting time slots. The fourth compares content types across several performance measures. The fifth shows the count of posts across platforms and content types. The project demonstrates how visual analytics can help users explore audience engagement, platform standing, and content patterns.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Social media analytics applies data analysis and visualization to engagement records, audience reactions, and content statistics. Social platforms generate detailed per-post data every day, which makes them well suited to analytics. This project uses a dataset of social media post performance and a Power BI report to summarize category-wise engagement, platform engagement, viral posts, and content-type comparisons.

1.2 Problem Statement

Social media datasets contain many rows and attributes. Reviewing them manually makes it difficult to compare platforms, content types, categories, and posting times. A visual analytics solution helps summarize records and present comparisons clearly.

1.3 Objectives

- Explore the social media engagement dataset and its attributes.
- Compare category-wise totals of likes, comments, shares, and engagement score.
- Compare likes and shares across the dataset.
- Examine how viral posts relate to impressions across posting time slots.
- Compare platforms by engagement score and examine platform and content-type composition.
- Compare content types across multiple performance measures.
- Build a multi-page Power BI report.

1.4 Scope of the Project

The analysis covers 3,000 records, 498 distinct user IDs (IDs range from 1 to 500), 6 platforms, 5 content types, 5 post categories, and 5 time slots as represented in the supplied dataset. Results describe this dataset and are not a live social media feed.

CHAPTER 2: TOOLS AND TECHNOLOGIES USED

2.1 Microsoft Power BI

Power BI is used to create interactive report pages and visualizations. The supplied PBIX file includes clustered column charts, a bar chart, a scatter chart, pie charts, a 100% stacked area chart, a donut chart, and a text box title.

2.2 Tabular Dataset

The data is held in a single table named social_media_analytics_dataset, imported into the Power BI data model. Such tabular data can also be stored as CSV and inspected using spreadsheet software, Python libraries, and Power BI.

2.3 Python and Pandas

Python and Pandas can be used to inspect the dataset, check data types, detect missing or duplicate rows, and calculate summary statistics.

2.4 Technologies Summary

Technology	Purpose
Tabular dataset (CSV)	Stores post-level data in rows and columns.
Microsoft Power BI	Creates charts and report pages.
Python	Supports data inspection and preprocessing.
Pandas	Loads and summarizes the dataset.

CHAPTER 3: DATASET DESCRIPTION

3.1 Dataset Overview

The Power BI data model contains one table, social_media_analytics_dataset, with 3,000 rows and 16 columns. The dataset includes post and user identifiers, platform and content information, posting time slot, engagement statistics, impressions, engagement rate, response time, sentiment score, engagement score, sentiment label, and a viral post flag.

3.2 Important Attributes

Attribute	Description
post_id	Unique identifier for each record (1 to 3,000).
user_id	Identifier for a user account (1 to 500).
platform	Platform of the post: Facebook, Instagram, LinkedIn, Snapchat, X (Twitter), or YouTube.
content_type	Format of the post: Image, Reel, Story, Text, or Video.
post_category	Subject of the post: Education, Entertainment, Lifestyle, News, or Promotion.
time_slot	Posting time: Morning, Afternoon, Evening, Night, or Late Night.
likes / comments / shares	Engagement statistics recorded for the post.
impressions	Number of times the post was displayed to users.
engagement_rate	Interactions as a percentage of impressions.
response_time_min	Time taken to respond to audience interaction, in minutes.
sentiment_score	Audience sentiment measure between -1 and 1.
engagement_score	Overall engagement score for the record (weighted interactions per 1,000 impressions).
sentiment_label	Sentiment class: Positive, Neutral, or Negative.
viral_post	Whether the post went viral (Yes or No).

3.3 Dataset Quality Checks

Basic inspection found 0 missing cells and 0 fully duplicated rows in the data table. These checks do not guarantee that every value is semantically valid. In particular, sentiment_label contains 2,135 Positive records against 653 Neutral and 212 Negative records, a skewed distribution that should be considered when interpreting results.

CHAPTER 4: METHODOLOGY

4.1 Project Workflow

Dataset → Data Inspection → Data Preparation → Power BI Import → Visualizations → Interpretation

4.2 Data Inspection and Preparation

1. Load the dataset and inspect rows and columns.
2. Check column names, data types, missing values, and duplicate records.
3. Confirm platform, content type, category, time slot, sentiment, and viral fields.
4. Aggregate records by category, platform, content type, time slot, sentiment, and viral flag.
5. Build report visuals in Power BI.

4.3 Dashboard Development

The supplied PBIX file contains five report pages. The visuals use sums of numeric fields (such as likes, comments, shares, impressions, and engagement score) and counts of records, grouped by category, platform, content type, time slot, sentiment label, and viral flag.

4.4 Interpretation

The visuals compare category-level totals, likes versus shares, platform engagement score, viral share of impressions by time slot, content-type totals, and the share of posts by platform and content type.

CHAPTER 5: POWER BI DASHBOARD AND VISUALIZATIONS

Dashboard Overview

The report contains five pages, each designed around a different analytical question. The charts below reproduce the visuals of each page from the supplied data.

5.1 Page 1: Engagement Overview

The first page carries the title “SOCIAL MEDIA ANALYTICS” and four visuals: a clustered column chart of category totals, a scatter chart by viral post flag, a pie chart of comments and shares, and a clustered column chart by sentiment label and category. Entertainment has the highest total engagement score (63,971.33) and Promotion the lowest (36,117.10). Entertainment has the most records (786) and the highest totals for comments (34,214) and shares (48,122). Shares total 188,684 against 135,380 comments, so shares make up 58.2% of the two combined.

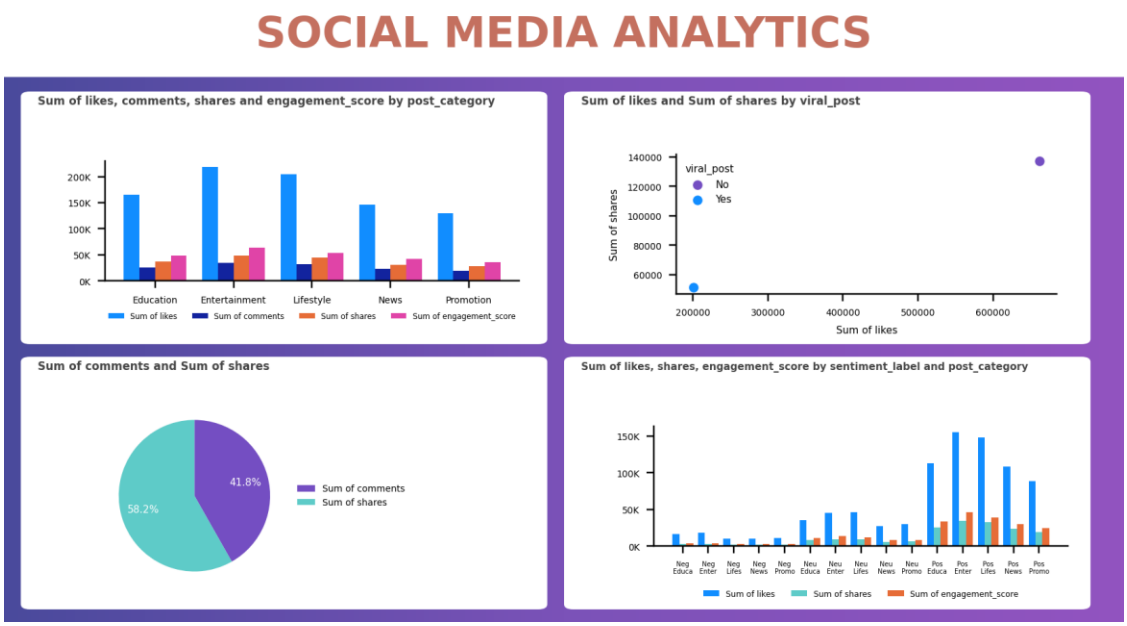


Figure 5.1: Page 1 – Engagement Overview

5.2 Page 2: Platform Engagement Score

The second page is a pie chart of summed engagement score by platform. Instagram has the largest share (58,902.01, or 24.09%), followed by YouTube (45,825.65) and Facebook (41,963.54). Snapchat has the smallest share (30,397.43, or 12.43%). Because these are sums, a platform with more records contributes more; Instagram has 670 records and Snapchat 380. By average engagement score, Instagram is also highest (87.91).

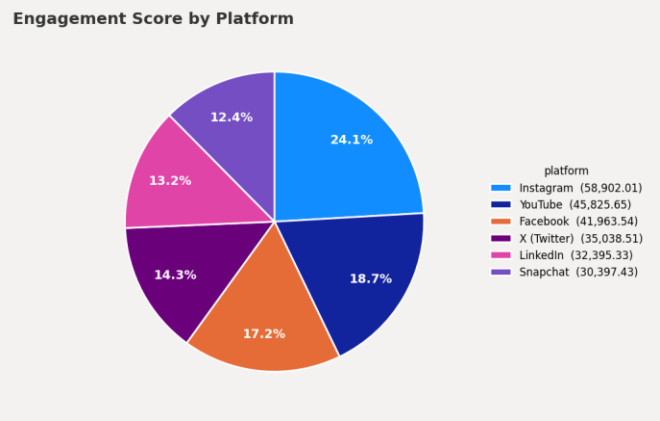


Figure 5.2: Page 2 – Engagement Score by Platform

5.3 Page 3: Viral Share by Time Slot

The third page is a 100% stacked area chart of impressions by time slot, split by viral post flag. Viral posts account for 23.7% of total impressions at Night, 21.2% in Late Night, 18.2% in the Evening, 17.9% in the Afternoon, and 15.6% in the Morning. The Evening slot also has the most viral posts (69).

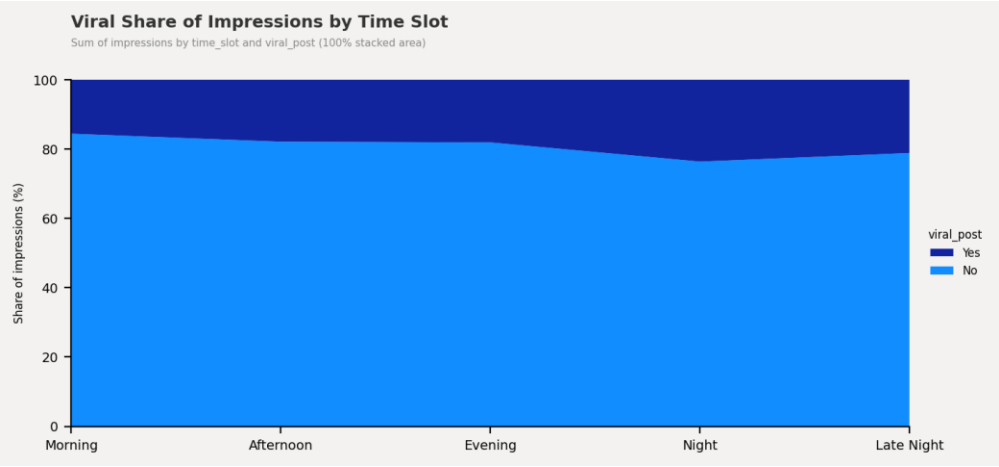


Figure 5.3: Page 3 – Viral Share of Impressions by Time Slot

5.4 Page 4: Content Type Comparison

The fourth page is a bar chart that compares content types on summed likes, comments, shares, engagement score, sentiment score, and response time. Reel has the highest totals on most measures (engagement score 64,551.84 and likes 310,615), while Text has the lowest. Part of this difference reflects record counts, since Reel has the most records (733) and Text the fewest (418).

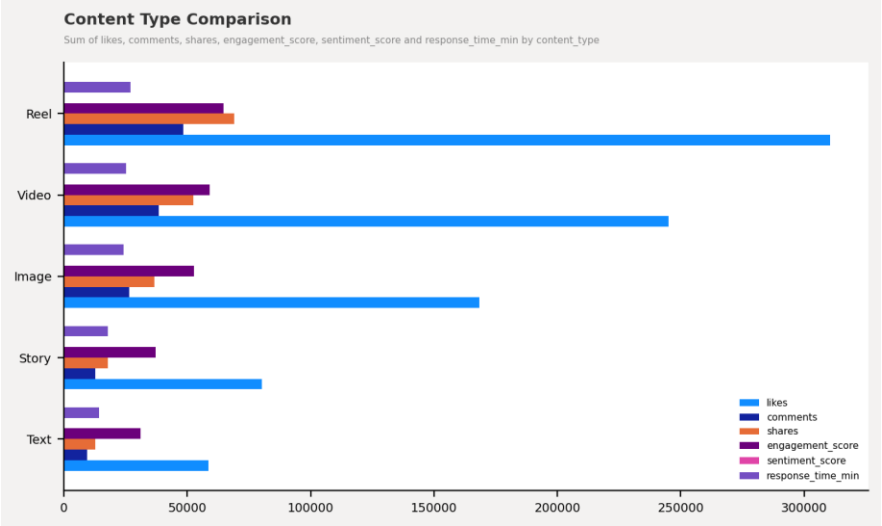


Figure 5.4: Page 4 – Content Type Comparison

5.5 Page 5: Post Count by Platform and Content Type

The fifth page is a donut chart of the count of posts by platform and content type, with a total of 3,000 posts. Instagram has the largest platform share (22.3%) and Snapchat the smallest (12.7%). The largest single platform-content segment is Instagram Reel (6.2%) and the smallest is LinkedIn Text (1.77%). For example, Instagram has 670 records and Snapchat 380.

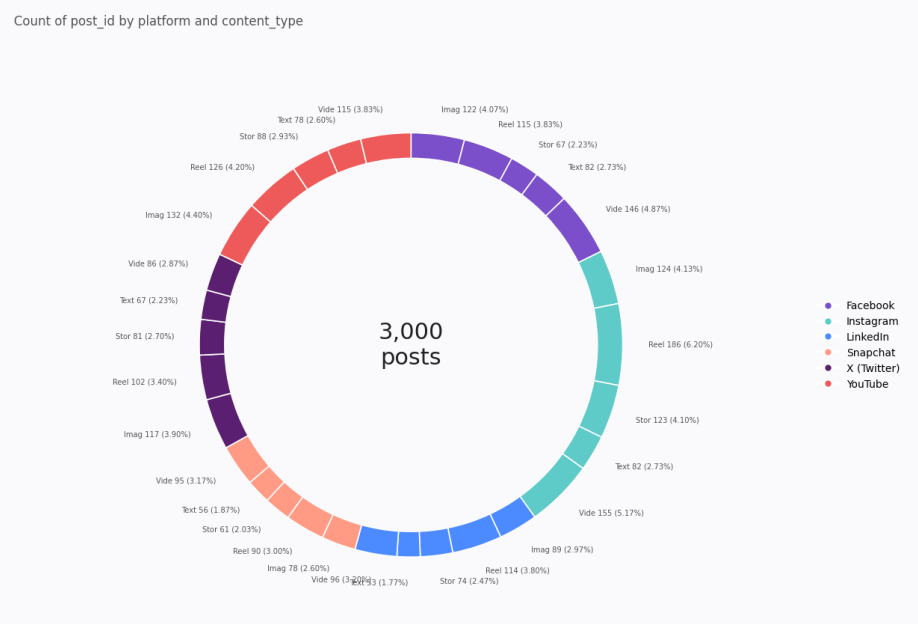


Figure 5.5: Page 5 – Count of Posts by Platform and Content Type

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Results

- The dataset contains 3,000 records covering 498 distinct user IDs, 6 platforms, and 5 content types.
- Sentiment labels are 2,135 Positive, 653 Neutral, and 212 Negative records.
- 242 records (8.1%) are flagged as viral posts.
- Instagram has the highest summed engagement score (58,902.01); Snapchat has the lowest (30,397.43).
- Entertainment has the highest total engagement score (63,971.33); Promotion has the lowest (36,117.10).
- Likes total 862,852, comments total 135,380, and shares total 188,684.
- Viral posts average an engagement score of 112.54, compared with 78.78 for non-viral posts.
- Reel has the most records (733) and Reel has the highest totals on most content-type measures.

6.2 Discussion

The dashboard combines category, platform, content-type, time-slot, and viral post information. These findings are descriptive. Most visuals use summed values, so groups with more records tend to show larger totals; averages or percentages would give a fairer comparison. The findings do not establish why a post performed well or whether any attribute caused higher engagement.

CHAPTER 7: ADVANTAGES, LIMITATIONS & FUTURE ENHANCEMENTS

7.1 Advantages

- Summarizes post-level records visually.
- Makes platform, content-type, and category comparisons easier.
- Shows how viral posts relate to impressions across posting times.
- Highlights platform and content-type composition.
- Provides a multi-page report in Power BI.

7.2 Limitations

- Results depend on the supplied dataset's correctness.
- Sentiment labels are skewed (2,135 Positive against 212 Negative), limiting comparisons across sentiment classes.
- Several visuals sum fields such as sentiment_score and response_time_min, which are not meaningful as totals.
- Sums are influenced by the number of records in each group; averages would be fairer.
- Counts alone do not explain the causes behind engagement.
- The dashboard is not live unless connected to an updating source.

7.3 Future Enhancements

- Add slicers for platform, content type, category, and time slot.
- Use averages and percentages (for example, average engagement rate and viral rate) instead of raw sums.
- Add a share-to-like ratio measure.
- Analyze the relationship between sentiment score, response time, and engagement score.
- Add drill-through pages for individual platforms or users.
- Validate the sentiment and viral post fields.

CHAPTER 8: DASHBOARD OUTPUT AND CONCLUSION

8.1 Dashboard Output

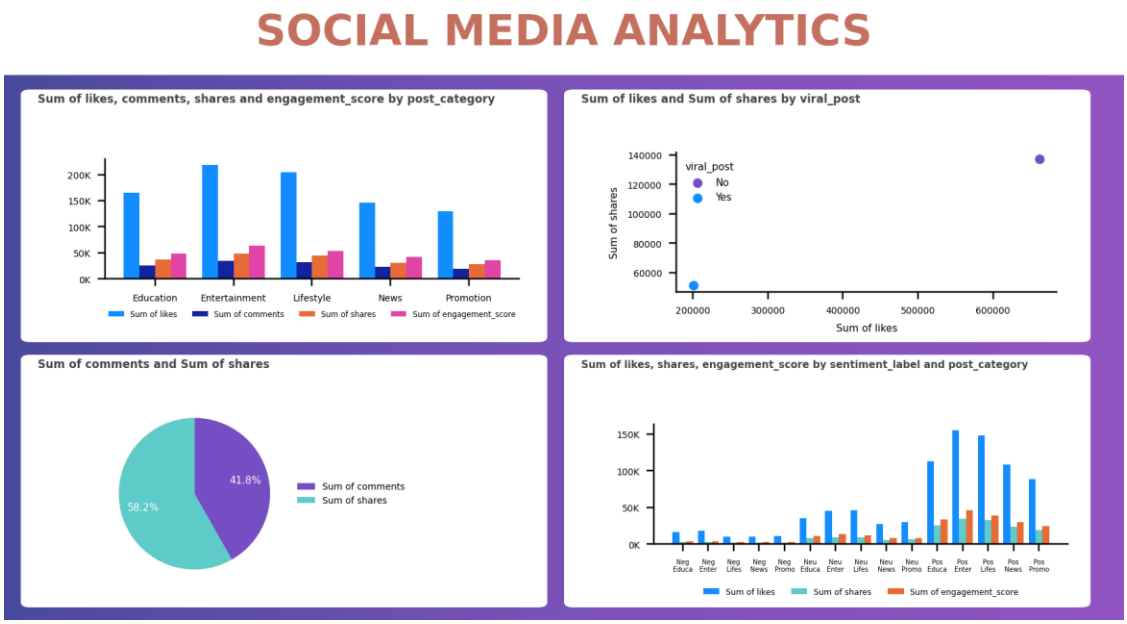


Figure 8.1: Power BI dashboard output (Page 1)

8.2 Conclusion

The Social Media Analysis project demonstrates how social media engagement data can be explored and summarized using Microsoft Power BI. The report presents category-level performance, likes versus shares, viral post distribution, platform engagement, content-type comparisons, and platform-content composition through visualizations.

The project provides practical experience in dataset inspection, aggregation, visual analytics, and dashboard presentation. Findings remain limited to the records and fields available in the supplied dataset.